

Introduction

One of the most important problems in data science is taking a very large dataset and making it (relatively) small. Doing this task well saves time and space and therefore also lots of money. Specifically, reducing the number of variables in a dataset is called dimension reduction (since the dimension of a data set is the number of variables it has). A good dimension reduction algorithm reduces the number of variables in a data set as much as possible (prioritizing space) while also maintaining the integrity of the data (prioritizing information). To balance these two goals well, data scientists might perform dimension estimation, which determines the number of variables that will attain the right equilibrium between space and information.

The most well-known method of dimension estimation is a linear algebra method called principal component analysis (which can also be used to do dimension reduction). However, in 2004, Elizaveta Levina and Peter J. Bickel [2] found a statistical method for dimension estimation using a maximum likelihood estimator. Over the course of this paper, we'll lay the theoretical foundations of dimension estimation, define some important statistical concepts, derive Levina and Bickel's MLE of intrinsic dimension, apply both the MLE method and PCA to simulated data, and compare the results of the two methods. Overall, we aim to both give an accessible introduction to the MLE method for intrinsic dimension estimation and determine how it compares to other leading methods in the field.

Theory of Intrinsic Dimension Estimation

Formally, a manifold is a subset of $\mathbb{R}^D$ that locally looks like $\mathbb{R}^d$, for $d < D$. For example, the Earth is a subset of 3-dimensional space, but for a long time humans assumed it to be flat (i.e. 2-dimensional) because that is how it appears locally.

A manifold has two types of dimensions: its ambient dimension and its intrinsic dimension. The ambient dimension of a manifold is the dimension of the space it is embedded in. For instance, if we draw a circle in a plane, the ambient dimension of the circle is 2. The intrinsic dimension of a manifold is the minimum number of parameters necessary to describe it. Returning to the circle example, we could represent the position of a point on the circle using two coordinates, x and y , which represent the point's position in relation to the two axes of the plane. Alternatively, we could use a single parameter θ which represents the point's position in terms of angle. Thus the intrinsic dimension of the circle is 1. A given manifold may have multiple potential values for its ambient dimension, but it will always have the same intrinsic dimension.

A common problem in data science is the task of dimension reduction, that is, representing the information captured by a real world data set with a new, smaller set of variables. Reducing the dimension of data is storage efficient and also makes tasks like training machine learning models quicker. Many popular dimension reduction algorithms (see, for example, [6]) require the user to input what dimension they would like to reduce their data to. This creates a need for an algorithm that can determine this target dimension.

The Manifold Hypothesis states that all real world data lies on or near a manifold of lower intrinsic dimension. Thus, given a data set, by determining the intrinsic dimension of the underlying manifold, one can determine the minimum number of variables necessary to represent the data. This process, called intrinsic dimension estimation, provides the

target dimension that is input into dimension reduction algorithms.

There are many approaches to intrinsic dimension estimation. This paper will describe one such approach, known as the Maximum Likelihood Estimator of Intrinsic Dimension. By defining a Poisson process on the observations in a data set, and treating them as embeddings of lower dimensional vectors, this method allows us to find a maximum likelihood estimator of the underlying intrinsic dimension of the data. The framework of finding MLEs given in STAT 311 will be useful in this approach, but the derivation will also require the reader to gain an understanding of new topics, such as inhomogeneous point processes.

Point Processes

The derivation of the MLE of intrinsic dimension will require a more complete knowledge of point processes than STAT 311 students have been exposed to so far. A point process is a “random collection of points falling in some space” [4]. Often this space is time, and point processes are used to model the arrival of events over time. However this space could also be some Euclidean space, such as the real line. In general, a point process is homogeneous if it has fixed rate over the whole space. An inhomogeneous process has rate that varies according to some variable t .

A binomial process is a type of point process, defined as follows. Given some random variables $X_1, X_2, \dots, X_n \stackrel{\text{iid}}{\sim} G$ and some set A , where G is a distribution with pdf g ,

$$N(A) = \sum_{i=1}^n \mathbb{1}_{\{X_i \in A\}}$$

is a binomial process. This process is called binomial because $N(A) \sim \text{Binomial}(n, g(A))$. An inhomogeneous Poisson process is one where the set A varies according to some t . Recall the definition of a Poisson process from Blitzstein and Hwang’s Introduction to Probability textbook [1].

A process of arrivals in continuous time is called Poisson with rate λ if:

1. The number of arrivals that occur in an interval of length t is a $\text{Poisson}(\lambda t)$ random variable.
2. The number of arrivals that occur in disjoint intervals are independent of each other

Students in Middlebury’s STAT 310 and 311 courses will have only encountered homogeneous Poisson processes thus far. An inhomogeneous Poisson process is very similar, except for the fact that the rate parameter, called $\lambda(t)$ varies with t .

Deriving an MLE of Intrinsic Dimension

First we’ll formally state the problem of dimension estimation. Suppose we are given some data $X_1, \dots, X_n$ which are i.i.d. Each $X_i \in \mathbb{R}^p$ is an embedding of a lower dimensional $Y_i \in \mathbb{R}^m$ where m is unknown and $m \leq p$. The Y_i are sampled from an unknown density f . What is the MLE for m ?

We'll now give a derivation of this MLE, elaborating on but following the closely the derivation given by Levina and Bickel in the original paper. Fix a point $x \in \mathbb{R}^p$. We'll assume that for a small enough sphere $S_x(R)$ of radius R centered around x , the density function f is constant. Now consider the following inhomogeneous binomial process $N(t, x), 0 \leq t \leq R$, given by

$$N(t, x) = \sum_{i=1}^n \mathbb{1}_{\{X_i \in S_x(t)\}}.$$

If we temporarily drop the dependence on x , it is true that as $n \rightarrow \infty$ and $R \rightarrow 0$, the binomial process $N(t, x)$ converges to a Poisson process, $N(t)$. To determine the rate, $\lambda(t)$, of $N(t)$, we must determine the rate at which $N(t)$ grows with respect to t . The approximate number of points within $S_x(t)$ is the value of the density function f evaluated at x , times the volume of $S_x(t)$. The volume of the unit sphere in m dimensions is

$$V(m) = \frac{\pi^{\frac{m}{2}}}{\Gamma(\frac{m}{2} + 1)}.$$

Then the volume of a sphere of radius t in m dimensions is simply $V(m) \cdot t^m$. So the approximate number of points within $S_x(t)$ is $f(x) \cdot V(m) \cdot t^m$. The rate at which this value grows is simply its derivative with respect to t . So $\lambda(t) = f(x) \cdot V(m) \cdot mt^{m-1}$.

The pdf of this inhomogeneous Poisson process given the lower dimension m and the value of the density function f evaluated at x can be written as

$$p(X_1, \dots, X_n | m, f(x)) = \exp \left\{ \int_0^R \log(\lambda(t)) dN(t) - \int_0^R \lambda(t) dt \right\}$$

[5]. Then the log-likelihood of m and $\theta = \log(f(x))$ given the data $X_1, \dots, X_n$ is

$$L(m, \theta | X_1, \dots, X_n) = \int_0^R \log(\lambda(t)) dN(t) - \int_0^R \lambda(t) dt.$$

Now take the derivative of L with respect to each of m and θ . To do this, first observe that

$$\begin{aligned} L(m, \theta | X_1, \dots, X_n) &= \int_0^R \log(f(x)V(m)mt^{m-1}) dN(t) - [f(x)V(m)t^m]_0^R \\ &= \int_0^R \log(f(x)) dN(t) + \int_0^R \log(V(m)mt^{m-1}) dN(t) - f(x)V(m)R^m \\ &= \int_0^R \log(f(x)) dN(t) + \int_0^R \log(V(m)mt^{m-1}) dN(t) - e^{\log f(x)} V(m) R^m \\ &= \int_0^R \log(f(x)) dN(t) + \int_0^R \log(V(m)mt^{m-1}) dN(t) - e^{\theta} V(m) R^m. \end{aligned}$$

Then

$$\frac{dL}{d\theta} = \int_0^R dN(t) - e^{\theta} V(m) R^m.$$

and

$$\frac{dL}{dm} = N(R) \left(\frac{V'(m)}{V(m)} + \frac{1}{m} \right) + \int_0^R \log t dN(t) - e^{\theta} V(m) R^m \left(\log R + \frac{V'(m)}{V(m)} \right).$$

We want the value of m such that L is maximized, so set both derivatives equal to 0 and solve for m . This gives us the following MLE for m :

$$\hat{m}_{MLE}(x) = \left[\frac{1}{N(R, x)} \sum_{j=1}^{N(R, x)} \log \frac{R}{T_j(x)} \right]^{-1}$$

where $T_j(x)$ is the Euclidean distance between x and its j th closest neighbor in $X_1, \dots, X_n$. When implementing the MLE for m , it is often easier to write it as follows:

$$\hat{m}_{MLE}(x) = \left[\frac{1}{k-1} \sum_{j=1}^{k-1} \log \frac{T_k(k)}{T_j(x)} \right]^{-1}$$

where k is the number of nearest neighbors. In practice, to obtain an estimate for m , we randomly select some number of points q and average over the estimates achieved at of the q points. So

$$\hat{m}_{MLE} = \frac{1}{q} \sum_{i=1}^q \hat{m}_{MLE}(x_i).$$

Application

Students who have taken Middlebury’s STAT 218 class have encountered principal component analysis as a method of dimension reduction. We seek to determine how the maximum likelihood estimator method performs against PCA. There are many ways to estimate intrinsic dimension using PCA. We will compare against the simplest method, often referred to as “max gap” which identifies the intrinsic dimension of the data as the location of the largest gap between the explained variances of the principal components. We wrote our own implementation of the maximum likelihood estimator method and used R’s built in “prcomp” function for PCA.

We will consider three types of manifold data: points uniformly sampled from the 2-sphere in 3 dimensions, points uniformly sampled from the 3-sphere in 4 dimensions, and points uniformly sampled from the swiss roll in 3 dimensions. To simulate the performance of this method on real-world data we will also consider a dataset of points uniformly sampled from the 2-sphere in 3 dimensions with a small amount of added Gaussian noise. The uniformly distributed points from the 2-sphere were obtained using a method called sampling by rejection [3].

Our MLE implementation requires the user to set the number of iterations to average over as well as the number of nearest neighbors to use. For each of the algorithms, we performed 50 iterations. To identify the appropriate k -value (the number of nearest neighbors), we plotted the results of the MLE method using from 2 nearest neighbors up to 40 nearest neighbors (Figure 1), and selected the value of k at which the results visually appeared to converge. We did this for each of the four data sets. For the three data sets without added noise, we saw convergence after about 20 nearest neighbors. However, the data with noise required at least 35 nearest neighbors.

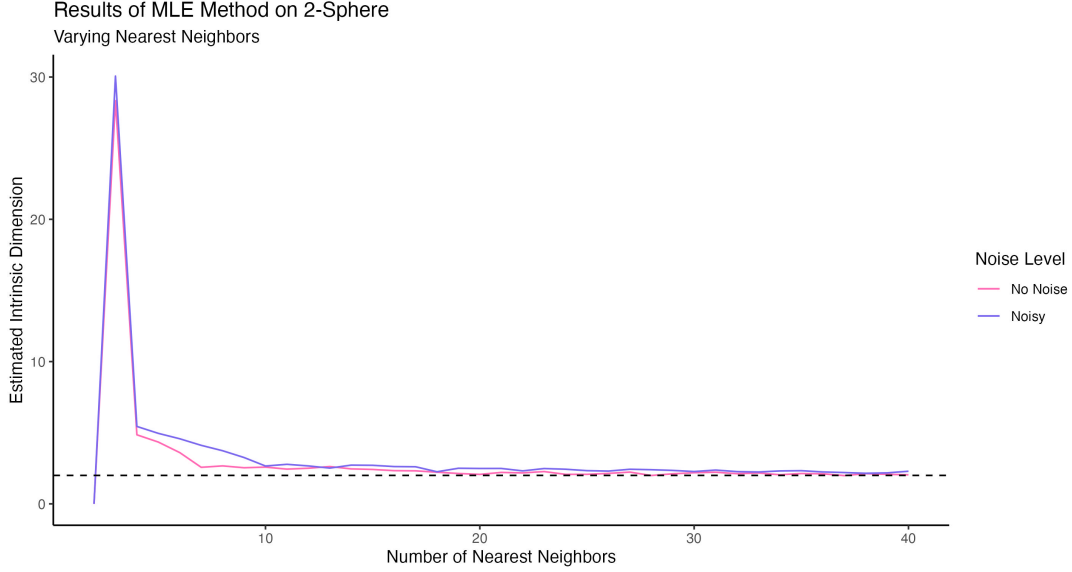


Figure 1

Using the k -values determined in the previous step, we obtained estimates of the intrinsic dimension of each of our data sets using the MLE and using PCA. The results are displayed in Figure 2. Assuming that the user of the MLE algorithm rounds the output to the nearest integer, the MLE outperformed PCA on all of the data.

Intrinsic Dimension Estimates by Method and Dataset

Data	Intrinsic Dimension	Ambient Dimension	ID Estimate MLE	ID Estimate PCA
2-Sphere	2	3	2.31	3
Noisy 2-Sphere	2	3	2.41	3
Swiss Roll	2	3	2.13	1
3-Sphere	3	4	3.33	4

Figure 2

Discussion

In STAT 218, students learn that PCA is a useful application of linear algebra techniques to reduce the dimension of data. However, PCA depends on linear relationships between data points and therefore has been shown to fail on nonlinear data [7]. This likely explains why PCA was unable to correctly identify the intrinsic dimension of our datasets, since the points on each of the manifolds we sampled from are nonlinearly related. Meanwhile, the MLE was able to successfully identify the intrinsic dimension of each of the data sets. Thus the maximum likelihood estimator of intrinsic dimension allows us to use STAT 0310 and 0311 topics to improve upon the methods learned in STAT 0218.

We also noted that when using the MLE method, we required a higher number of nearest neighbors to obtain an estimate for data with added noise. In future work, one might do a more rigorous mathematical analysis of the impact on the estimate of adding a normally distributed random variable to each of the data points.

It is important to recognize that our study was conducted only on datasets of relatively low ambient and intrinsic dimension. While this allowed for simpler visualization and

sampling, it only demonstrated the performance of the MLE on a certain type of data. Since dimension estimation methods are intended to be used on data of high ambient dimension, it would be valuable to carry out this comparison on such high-dimensional data, particularly on real world data. The MLE of intrinsic dimension has been shown to be asymptotically biased towards lower intrinsic dimensions [2]. This bias did not cause the MLE to perform poorly in our study, but an expansion of the analysis to higher-dimensional data would help to determine whether this bias impacts the performance of the estimator in practice.

References

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